Week 3

Week 3 Status Update — Phase 3

Phase 3: RAG Reading Generation

- Built a **Docker Container Lambda** that performs RAG-based reading generation
- Successfully tested two input modes:
  - **Query only** → FAISS search → Claude generates reading (e.g., "What is a Generator type?")
  - **chart_data + query** → constructs natural language query from chart properties (chart_data (json) + prompt) → FAISS search → (chart_data + chunks + query) → Claude generates personalized response

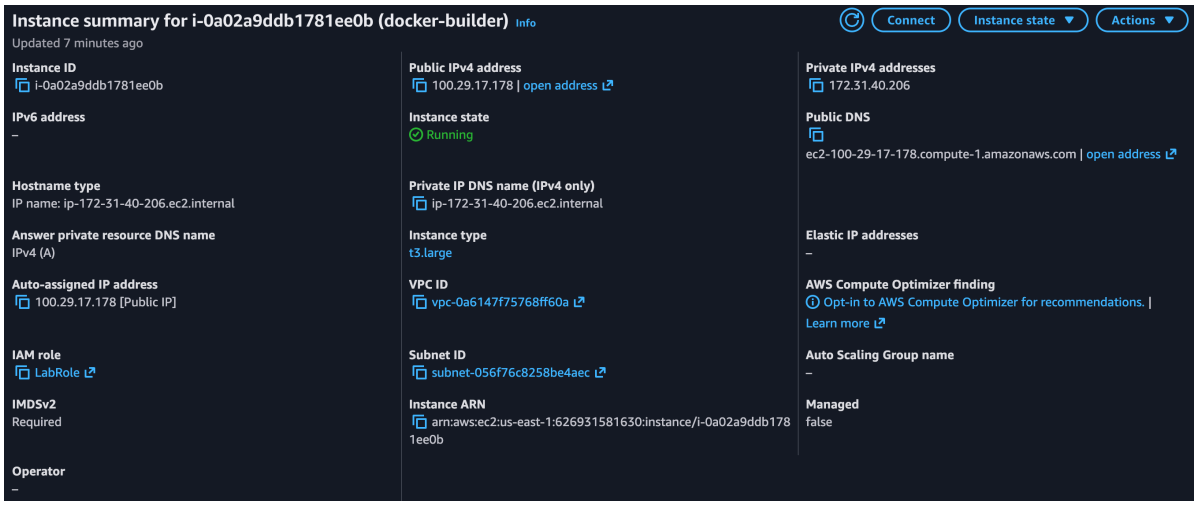
Technical challenges solved

1. **Lambda ZIP size limit (250MB)**: sentence-transformers + PyTorch + faiss-cpu totals over 1GB. Solved by using Docker Container deployment (10GB limit) instead of ZIP.

2. **Read-only file system error**: sentence-transformers attempted to download the model to the home directory at runtime, but Lambda's filesystem is read-only except `/tmp`. Solved by pre-downloading the embedding model into the Docker image at build time (`/opt/models`), eliminating any runtime downloads.

Build and deployment flow

- Used **EC2** (t3.large) to build the Docker image (~8GB) containing all dependencies



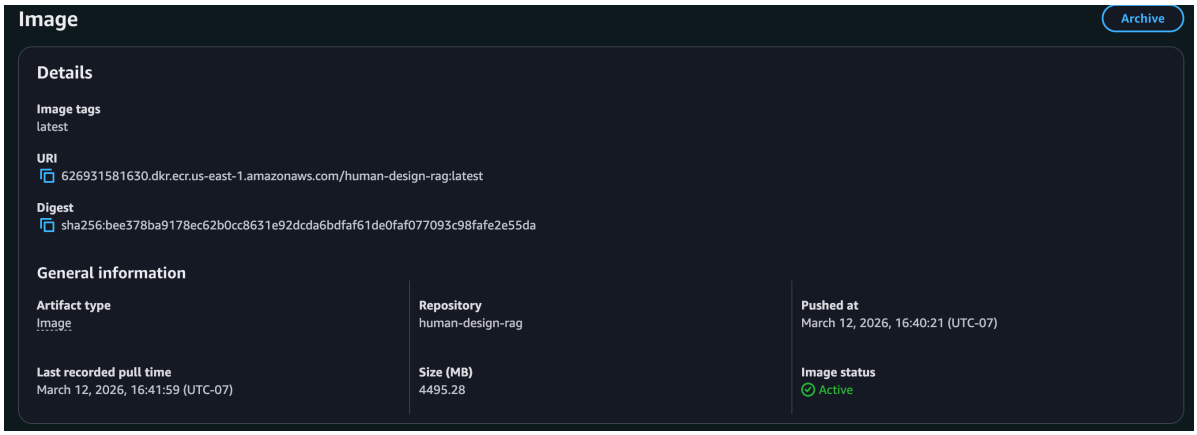
Connect to SSM Session Manager then upload the `.py` file and create Dockerfile for push to ECR.

```
[ec2-user@ip-172-31-40-206 lambda-rag]$ ls
Dockerfile  lambda_function.py
```

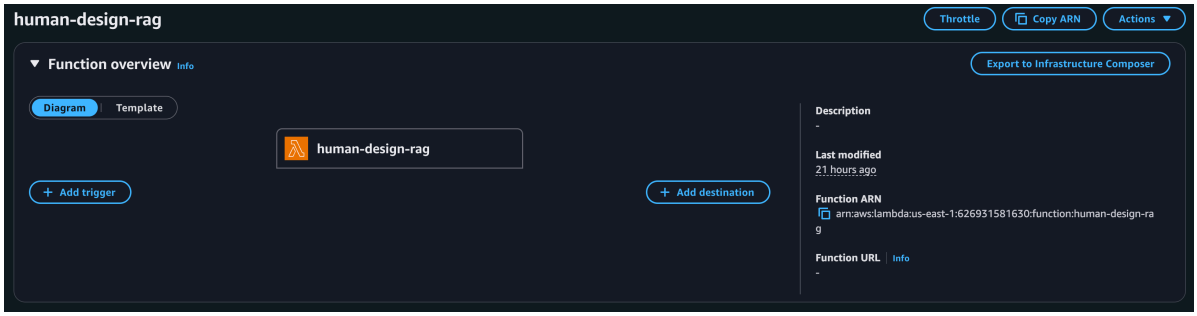
```
# Create a repository called human-design-rag in ECR
# This is where the Docker Image is stored — Lambda will p
ull the image from here
aws ecr create-repository --repository-name human-design-r
ag --region us-east-1

# Log in to ECR so Docker can push images to it
aws ecr get-login-password --region us-east-1 | docker log
in --username AWS --password-stdin 626931581630.dkr.ecr.us
```

```
-east-1.amazonaws.com

# build docker image
docker build -t human-design-rag .
#  => => naming to docker.io/library/human-design-rag

# Tag the local image with the full ECR address so Docker
knows where to push it
docker tag human-design-rag:latest 626931581630.dkr.ecr.us
-east-1.amazonaws.com/human-design-rag:latest

# push to ECR
docker push 626931581630.dkr.ecr.us-east-1.amazonaws.com/h
uman-design-rag:latest
```

- Pushed image to **ECR** (Elastic Container Registry)



- Created Lambda function pointing to the ECR image with `LabRole`



- Configured Lambda: 3008MB memory, 5-minute timeout, environment variables for `ANTHROPIC_API_KEY` and `BUCKET_NAME`

Lambda execution flow

1. Cold start: downloads `faiss_index.bin` and `chunks.json` from S3 to `/tmp`
2. Loads FAISS index and pre-packaged sentence-transformers model into memory
3. Converts user query (or chart_data-derived query) into a vector
4. FAISS retrieves **top 7** most relevant book passages
5. Sends chart_data + retrieved passages + user question to Anthropic Claude Sonnet API
6. Claude generates a personalized reading using only the provided passages (prompt enforces no hallucination)

Test results

- **Query only test**: Successfully returned a comprehensive Generator type explanation, sourced from *The Definitive Book of Human Design*
- **chart_data + query test**: Successfully returned a personalized career reading for a Generator with Sacral authority, profile 4/6, and Channel 34-20, referencing both source books

```
# 1. query only
{
  "body": "{\"query\": \"What is a Generator type in Human De
sign?\"}"
}
# import PyTorch and download FAISS index from S3
{
  "statusCode": 200,
  "headers": {
    "Content-Type": "application/json"
  },
  "body": "{\"success\": true, \"reading\": \"Based on the re
```

```
ference passages and your Human Design chart, let me explain
what it means to be a Generator type.\\n\\n## What is a Gener
ator Type?\\n\\nAs a Generator, you are part of the largest g
roup in humanity - Generators (including pure Generators and
Manifesting Generators) make up about 70% of all people. Pure
Generators specifically comprise approximately 37% of humanit
y.\\n\\n## Your Core Design\\n\\n**Sacral Energy:** Your defi
ning characteristic is having a defined Sacral Center, which
is your source of sustainable life-force energy. This sacred
creative energy is what drives you and gives you your remarka
ble stamina and endurance.\\n\\n**Response-Based Living:** Un
like Manifestors who can initiate, you are designed to respon
d rather than initiate. Your strategy is to wait for life to
come to you and then respond with your Sacral energy. This re
sponse comes as sounds or vibrations that rise from your mids
ection to your throat, indicating whether your Sacral has ene
rgy available for what's being presented.\\n\\n## How You Ope
rate\\n\\n**Energy Management:** You have a daily supply of S
acral energy that needs to be used properly. When you respond
correctly to opportunities, work, and requests, you engage in
activities that truly satisfy you and transform your life.\\n
\\n**Step-by-Step Process:** Once you respond and engage in a
n activity, you naturally go through a thorough, step-by-step
process that covers each stage completely. You cannot quickly
change gears once committed to a task or direction.\\n\\n**Wo
rk and Life Integration:** You find your life through your wo
rk and how you use your daily energy supply. The right work a
ctually transforms your life and can slow the degenerative pr
ocess in your body.\\n\\n## Your Challenge and Growth\\n\\n**
Letting Go of Initiation:** One of your biggest challenges is
accepting that you cannot manifest your own ideas for your li
fe. You need to release mental pictures of what your life
\\\"should look like\\\" and avoid forcing your dreams into r
eality through initiation.\\n\\n**Frustration as Your Guide:*
* When you try to initiate or push things to happen, you meet
resistance and experience frustration. This frustration is ac
```

tually your signpost to return to your responsive nature.\\n
\\n**Satisfaction Through Response:** Your fulfillment comes
from responding correctly to what life brings you, rather tha
n trying to make things happen on your own.\\n\\n## Your Myst
ical Path\\n\\nYour spiritual journey involves waking up your
Sacral center and learning to live surrendered to its subject
ive truth. This means trusting your gut responses and honorin
g the wisdom of your body over the desires and fantasies of y
our mind.\\n\\nAs a Generator, you are here to be the sustain
able creative life-force energy of humanity, but only when yo
u honor your responsive nature and follow your Sacral guidanc
e.\", \"sources\": [{\"source\": \"pdfcoffee.com_human-design
-the-definitive-book-of-human-design-the-science-of-pdf-free.
pdf\", \"score\": 0.32750988006591797}, {\"source\": \"pdfcof
fee.com_human-design-the-definitive-book-of-human-design-the-
science-of-pdf-free.pdf\", \"score\": 0.2504398822784424},
{\"source\": \"pdfcoffee.com_human-design-the-definitive-book
-of-human-design-the-science-of-pdf-free.pdf\", \"score\": 0.
20366811752319336}, {\"source\": \"pdfcoffee.com_human-design
-the-definitive-book-of-human-design-the-science-of-pdf-free.
pdf\", \"score\": 0.17929482460021973}, {\"source\": \"pdfcof
fee.com_human-design-the-definitive-book-of-human-design-the-
science-of-pdf-free.pdf\", \"score\": 0.17876899242401123},
{\"source\": \"pdfcoffee.com_human-design-the-definitive-book
-of-human-design-the-science-of-pdf-free.pdf\", \"score\": 0.
15862411260604858}, {\"source\": \"pdfcoffee.com_human-design
-the-definitive-book-of-human-design-the-science-of-pdf-free.
pdf\", \"score\": 0.15519917011260986}]}"
}

# 2. query + image json
{
  "body": "{\"chart_data\": {\"type\": \"Generator\", \"autho
rity\": \"Sacral\", \"profile\": \"4/6\", \"defined_centers
\": [\"Sacral\", \"Throat\", \"Solar Plexus\"], \"active_chan

nels\": [\"34-20\"]}, \"query\": \"What does my type mean for my career?\"}"
}

{
  "statusCode": 200,
  "headers": {
    "Content-Type": "application/json"
  },
  "body": "{\"success\": true, \"reading\": \"# Your Human Design Type and Career Path\\n\\n## Understanding Your Generator Type\\n\\nAs a Generator with a defined Sacral Center, you are part of approximately 70% of the population and represent the creative life force energy on our planet. Your Sacral Center is your powerhouse - it generates the sustainable energy that fuels not just your own life, but contributes to the vitality of everyone around you.\\n\\n## Your Career Strategy: The Power of Response\\n\\n**The most crucial aspect for your career success is following your Generator Strategy: to respond.** The reference materials are very clear about this - your Sacral Center has a \\\"receptive frequency designed to respond to what it meets in a protective, healthy, honest, creative and persevering manner.\\\"\\n\\nThis means for your career:\\n- **Don't initiate career moves impulsively** - Generators \\\"meet with resistance when they try to initiate\\\"\\n- **Wait for opportunities to come to you, then respond** - Your Sacral will give you clear gut responses about what's right\\n- **Trust your Sacral Authority** - Your defined Sacral Center will generate \\\"uh-huh\\\" (yes) or \\\"uh-uh\\\" (no) responses to career opportunities\\n\\n## Your Unique Career Advantage: Channel 34-20\\n\\nYou have something special that sets you apart from pure Generators - **Channel 34-20 connecting your Sacral directly to your Throat Center.** According to the materials, this creates \\\"an anomaly where the Sacral's generative power is converted into a potential for manifestation.\\\"\\n\\nThis means you can:\\n- **Move from resp

onse to action more quickly** than other Generators\\n- **Manifest your ideas** once your Sacral has responded positively\\n- **Be a potential powerhouse** in your chosen field\\n\\n## Career Sustainability and Energy Management\\n\\nYour Sacral Center is designed for **work, reproduction, sexuality and perseverance.** For your career, this translates to:\\n\\n- **You're built for work** - but it must be work that your Sacral responds to positively\\n- **You have sustainable energy** when doing what you love\\n- **Burnout happens when you force things** - when your Sacral isn't truly engaged, \\\"its generative capacity quickly becomes degenerative and burnout results\\\"\\n\\n## Practical Career Guidance\\n\\n1. **Pay attention to your gut responses** when career opportunities arise\\n2. **Wait for invitations, job postings, or requests** rather than cold-calling or aggressive networking\\n3. **Once your Sacral says yes, you can move forward with confidence** - your Channel 34-20 gives you the power to manifest\\n4. **Choose work that engages your life force** - you'll have the stamina and perseverance for what truly lights you up\\n5. **Trust the process** - your \\\"built-in guidance system of the gut response\\\" will lead you to the right career path\\n\\nYour Generator nature means you're here to do the work that sustains life and civilization. When you follow your Strategy and Authority, you'll find yourself in a career that not only fulfills you but allows your considerable energy and resources to flow effectively into the world.\", \"sources\": [{\"source\": \"pdfcoffee.com_human-design-the-definitive-book-of-human-design-the-science-of-pdf-free.pdf\", \"score\": 0.17906606197357178}, {\"source\": \"pdfcoffee.com_human-design-the-definitive-book-of-human-design-the-science-of-pdf-free.pdf\", \"score\": 0.09558999538421631}, {\"source\": \"pdfcoffee.com_human-design-the-definitive-book-of-human-design-the-science-of-pdf-free.pdf\", \"score\": 0.09520775079727173}, {\"source\": \"Human Design_ Discover the Person You Were Born to Be - PDF Room.pdf\", \"score\": 0.09276819229125977}, {\"source\": \"pdfcoffee.com_human-design-the-definitive-book-of-human

```
-design-the-science-of-pdf-free.pdf\", \"score\": 0.076037049
29351807}, {\"source\": \"pdfcoffee.com_human-design-the-defi
nitive-book-of-human-design-the-science-of-pdf-free.pdf\",
\"score\": 0.07428503036499023}, {\"source\": \"pdfcoffee.com
_human-design-the-definitive-book-of-human-design-the-science
-of-pdf-free.pdf\", \"score\": 0.056825339794158936}]}"
}
```


- Cold start duration: ~190 seconds (includes S3 download + model loading + API call)
- Memory used: ~1089MB of 3008MB allocated

Services used: EC2 (build), ECR (image registry), Lambda (Container), S3, FAISS, sentence-transformers, Anthropic Claude Sonnet API

Next steps

- Phase 4: API Gateway endpoint + Streamlit frontend